

Volumetric Titration

Titration is routinely used in almost all areas of Chemistry. In a titration, the analyte that needs to be quantitatively determined is made to react with a reagent solution of known concentration usually added from a burette ("titrant"). Such a reagent solution is known as a standard solution. The volume of the standard solution required to be completely react with the analyte is measured and the amount of analyte can thereby be calculated. A titration should fulfill several requirements; so that it can be used in quantitative work.

- The reaction should be stoichiometric.
- The reaction should be rapid.
- There should be no side reactions and interfering substances.
- There should be a noticeable change in the solution when the reaction is complete, which can be indicated by the use of an indicator.
- The equivalence point of the reaction (i.e. when the analyte and reagent have reacted in stoichiometric amounts) should coincide with, or be as close as possible to the end point (i.e. when the change in the property measured in (d) above is detected).
- The reaction should be quantitative (i.e. the reaction should go to completion, or the equilibrium of the reaction should lie far to the right).

Standard solutions used in titrations are prepared by accurately weighing pure substances (known as primary standards) and dissolving them in a solvent in order to obtain a solution of accurate volume. If high purity substances are not available, then a solution of approximate concentration may be prepared and therefore standardized using primary standards. Primary standards should possess following requirements.

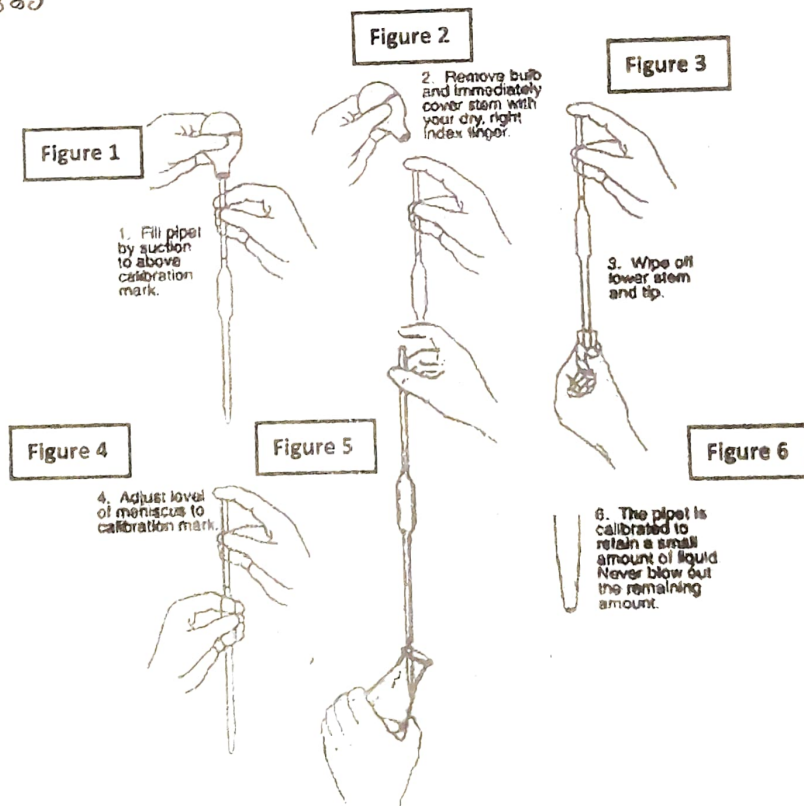
- 100% pure.
- Stable at room temperature and should also be stable at drying temperature.
- Readily available and cheap.
- Preferably, have high molecular weights in order to minimize errors when weighing.

Some Common Indicators

Indicator	Transition Range (pH)	Acid Color	Base Color
Methyl Orange	3.1 – 4.4	Red	Orange
Methyl Red	4.8 – 6.0	Red	Yellow
Bromothymol Blue	6.0 – 7.6	Yellow	Blue
Phenolphthalein	8.0 – 9.6	Colorless	Red

Measurement of aliquot using pipette

- Use a pipette filler to draw a small volume of the liquid to be sampled into the pipette.
- Thoroughly wet the entire interior surface.
- Repeat with at least two additional portion.
- Carefully fill the pipette to a level above the graduation mark. (Figure 1)
- Quickly replace the bulb with a forefinger to arrest the outflow of liquid (Figure 2). Make sure that there are no air bubbles in the bulk of the liquid or foam at the surface.
- Tilt the pipette slightly from the vertical and wipe the exterior free of adhering liquid (Figure 3).
- Touch the top of the pipette to the wall of a glass container (not the container into which the aliquot is to be transferred) and slowly allow the liquid level to drop by partially releasing the forefinger (Figure 4). Halt further flow as the bottom of the meniscus coincides exactly with the graduation mark.
- Place the pipette tip well within the receiving flask and allow the liquid to drain (Figure 5).
- When free flow ceases, rest the tip of the pipette, against the inner wall of the receiver for 10 seconds (Figure 6).
- Withdraw the pipette with a rotating motion to remove any liquid adhering to the tip.



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Note: Small volume retaining inside the tip of the pipette should not be blown or rinsed into the receiving flask.

Directions for the use of a burette

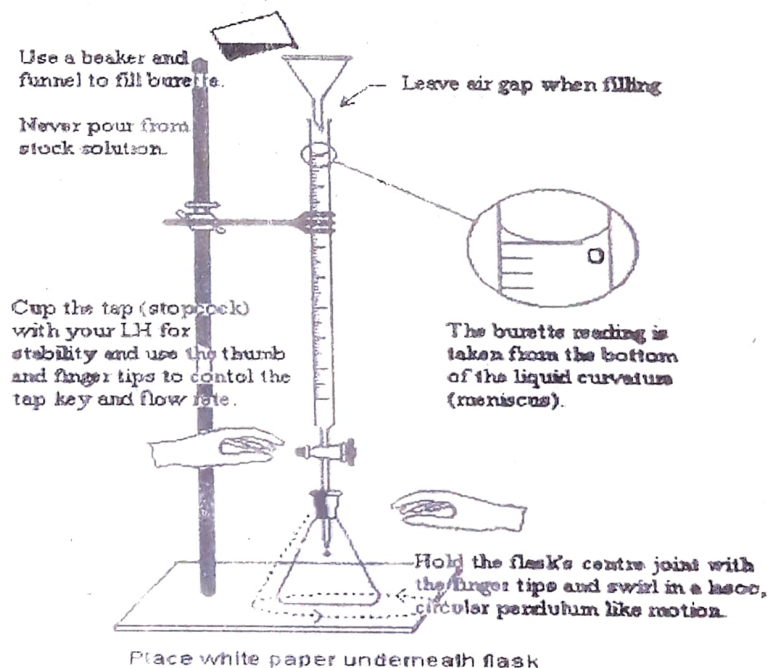
Test for leakages: Fill the burette with water and check whether volume reading changes with time.

Drying: Open the stopcock and clamp the burette in an inverted position.

Method for reading a burette: The eye should be level with the meniscus.

Filling:

- Close the stopcock and add 5-10 cm³ of the titrant.
- Then carefully rotate the burette to wet the interior completely.
- Allow the liquid to drain through the tip.
- Repeat this step at least two more times.
- Then fill the burette well above the zero mark.
- Remove air bubbles by rapidly rotating the stopcock and permitting small quantities of the titrant to pass.
- Lower the level of the liquid to the zero mark.
- Allow for drainage (~ 1 min) and then record the initial volume reading (estimating to the nearest 0.05 cm³).
- Figure 7 indicates the best method to handle the stopcock.



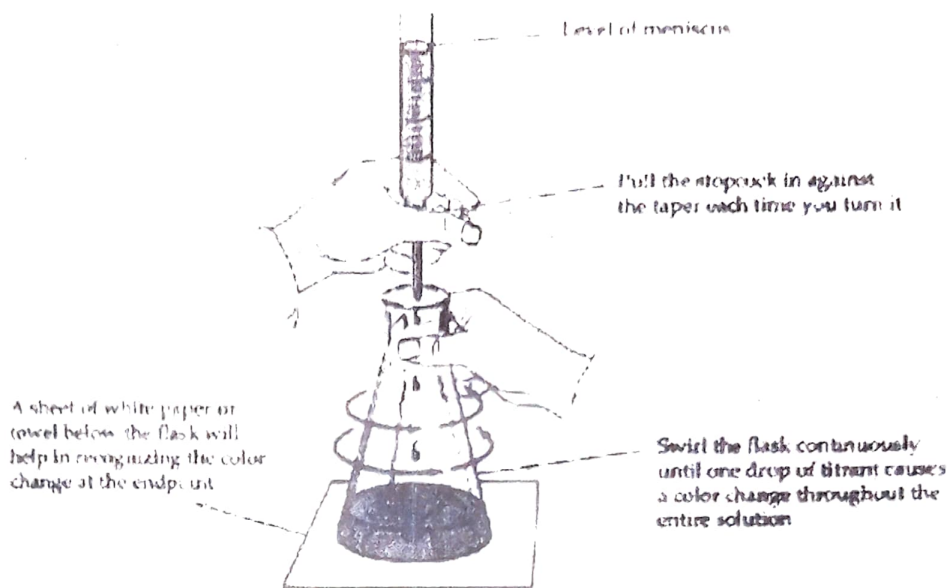
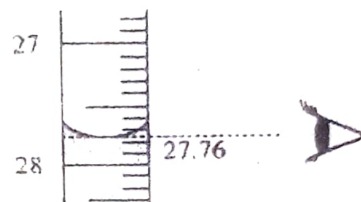
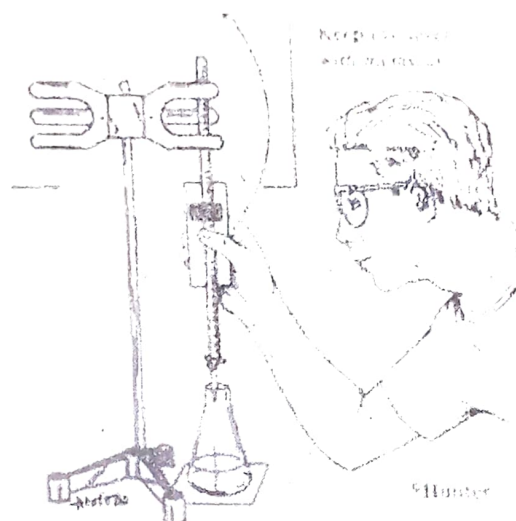


Figure 7

Titration Procedure

- Clean the burette before use and rinse with distilled water.
- Once the burette is clean, rinse it with about 5 mL of titrant solution before filling it.
- Place the burette vertically in a burette clamp attached to a stand.
- Using a funnel, fill the burette with titrant to a level above the zero mark.
- Remove the funnel and place a beaker under the burette and open the stopcock for a few second to remove all air from the tip and fill it again. Be sure your stopcock fits snugly and that the burette does not leak.
- Lower the level of the liquid and adjust to zero (0.00) by reading the burette with the meniscus level with the eye to minimize parallax. If necessary place a black strip behind the burette to make the meniscus easier to see and the volume easier to read.
- Draw a small volume of the solution to be titrated into the pipette and rinse it. Do not rinse the titration flask with the solution to be titrated.
- Carefully fill the pipette to a level above the graduation mark and slowly allow the liquid level to drop until the meniscus coincides exactly with the graduation mark (Figure 8).
- Transfer the solution into a clean dry titration or Erlenmeyer flask, add 3-5 drops of the appropriate indicator and position it on a white tile beneath the burette.
- Add the titrant slowly from the burette while swirling the content of the flask to assure adequate mixing.

- When the end point wash the wall of the flask down with a little distilled water from a wash bottle and continue the titration very carefully till the end point.
- When nearing the end point has been reached, allow the solution to sit for 10 seconds so the liquid in the burette can settle, then read the burette. Subtract the initial burette reading from the final reading to obtain the volume of titrant used.
- The procedure is repeated with two fresh portions of the titrant.
- All the burette readings are made to 0.05 mL. (eg; 25.75, 12.30, 00.00 mL)



Move your eyes to the level of the liquid surface and read the value of the bottom of the meniscus with 1/10 of the smallest scale marked on the burette.

Figure 8

ACID BASE TITRATIONS

The chemical reaction involved in acid- base titration is known as neutralization reaction. It involves the combination of H_3O^+ ions with OH^- ions to form water. In acid- base titrations, solutions of alkali are titrated against standard acid solutions. The estimation of an alkali solution using a standard acid solution is called acidimetry. Similarly, the estimation of an acid solution using a standard alkali solution is called alkalimetry.

Day ①

Experiment No: 01- Standardization of NaOH solution using standard HCl solution

Procedure:

Pipette out 25.00 mL of the NaOH solution into a titration flask and add 2 drops of Methyl Orange/ Phenolphthalein indicator. Titrate this solution against the standard HCl solution in the burette.

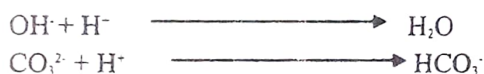
Experiment No: 04- (i) Titration of a mixture of CO_3^{2-} and OH^- using standard HCl solution

Theory:

This method utilizes a titration with a mixture of indicators to give the desired results. First, a portion of the solution containing both CO_3^{2-} and OH^- is titrated against standard HCl solution using Methyl Orange (pH range = 3.1- 4.4) as an indicator and suppose the volume of HCl required is V_1 mL. The end point corresponds to the following reactions.



Next another portion of the same solution is titrated against standard HCl solution using a mixture of the indicators; Thymol blue and Cresol red (6:1), and suppose the volume of HCl required now is V_2 mL. The end point corresponds to the following reactions.



Procedure:

Pipette out 25.00 mL of the solution containing both CO_3^{2-} and OH^- into a titration flask, add 2 drops of Methyl Orange. Titrate this against standard HCl solution in the burette until the color changes from yellow to red as before.

Next, titrate another 25.00 mL portion of the solution both CO_3^{2-} and OH^- with the standard HCl solution using Thymol blue and Cresol red as the indicator until the color changes from blue to pink.

(ii) Titration of a mixture of CO_3^{2-} and HCO_3^- using standard HCl solution

Procedure:

Pipette out 25.00 mL of the solution containing both CO_3^{2-} and OH^- into a titration flask and add a drop of Phenolphthalein as the indicator and titrate this against standard HCl solution in the burette until the color changes from pink to colorless.

Add 2 drops of Methyl Orange to another 25.00 mL of the solution "A" and carryout the titration until the color changes from yellow to orange.

Calculate the concentration of both CO_3^{2-} and HCO_3^- in the given solution "A".